

James Y. Wang

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EDUCATION

Georgia Institute of Technology

- B.S. in Mechanical Engineering Candidate
- Awards – *Faculty Honors, Dean's List, Tau Beta Pi Engineering Honor Society*

Atlanta, GA
GPA - 3.86 / 4.00
Expected Dec. 2027

WORK EXPERIENCE

McLaren Formula One Team

Engine Systems Design Intern

Woking, Surrey, UK
July 2025 - July 2026

- **Designed heat-treated linkage pins for active aero mechanisms** while satisfying fatigue and load constraints.
- **Redesigned high-pressure hydraulic manifold** to fit new connections by repositioning bores and internal features while meeting minimum wall thickness requirements; produced multi-sheet manufacturing drawings with extensive GD&T.
- **Developed remote-access steering rack bleed tool** with multiple sealing elements and ball check valve to bleed the rack externally; owned design, detail drawings, assembly drawings, and supplier sourcing.
- Designed composite fluid-tank internals with optimized positioning and cutouts to increase tank rigidity and reduce weight.
- Created titanium keep plates for hydraulic manifolds to retain high-pressure quick-disconnect connectors.
- Designed conformal bonded carbon brackets to route fuel and wire harnessing within engine bay internals.
- Developed hydraulic piston sleeve tooling and Lee plug press tooling blocks for actuators and hydraulic manifolds.

Bodywork Design Intern

- **Designed conformal front-wing sensor mounts and wire harnessing** used on-car in practice, including low-intrusion packaging, assembly/trimming drawings, and bonding jig design.
- Engineered 3-part mold, pattern, and laminates for structural rear wing cover panel.
- Developed patterns, molds, and laminate manuals for rear wing fairings to support new upgrades in close collaboration with aerodynamics and structures teams.

Strategy and Sporting Intern

- **Worked mission control on race weekends**, performing live pit loss calculations, driver pitlane performance analysis, team/competitor reliability tracking, and competitor analysis for live strategy decisions.
- Developed live pitstop telemetry website that provided live one-click pitstop analysis using on-car telemetry, while also being able to compare historical runs with any telemetry channel on the car.
- Attended Bahrain pre-season test to support pitstops by providing video/telemetry analysis and pit crew feedback to drivers and performance engineers to improve braking consistency, stopping position, and approach execution for faster pitstops.

Tesla

Model 3/Y Battery Module Process & Equipment Engineering Intern

Reno, NV
May 2024 - Aug. 2024

- **Achieved \$900k 2-year ROI by designing a fleet of motorized carts** capable of carrying 1 metric ton of batteries.
- Owned design and process of new coupler lubrication station with improved part tracking and assembly time.
- Trained and tuned computer vision quality-control models for battery module assembly line using custom image datasets.

SpaceX

Starship Mechanisms Engineering Intern

Hawthorne, CA
Jan. 2024 - May 2024

- **Designed multi-function rotary build stand for thrust vector actuators** and owned assembly, wiring, and sealing processes for Starship Flight 5; stand enabled multiple assembly stages and supported production for dozens of units.
- Designed jig and process to trim bar-wound electric motor stators for laser welding with a high-speed custom slit saw to alleviate weld quality and arc risk concerns due to embedded debris.
- Redesigned high-stress proofing tool used during thrust vector actuator assembly using hand calculations and FEA.

Boeing

777F/777X Tools and Equipment Engineering Intern

Everett, WA
May 2023 - Aug. 2023

- **Saved \$5500/19 rework hours by redesigning alignment jig** for the 777F in order to prevent damage to the fuselage using CATIA V4 and CATIA V5, 3D Printing, and GD&T.
- Designed multi-aircraft compatible machined reflectors to be used with laser projectors to aid locating rivet positions.
- Prevented 16 hours of lost time by designing a conforming foam kneeling pad to prevent injury in the 777 rear bulkhead.

PROJECT EXPERIENCE

Analog Headphone Project

Aug. 2024 - Dec. 2024

- Designed a custom PCB for 3D-printed headphones with adjustable gain, two 50mm speakers, and rechargeable battery.

Georgia Tech Experimental Rocketry (GTXR) - Structures/Externals Subsystem

Sept. 2023 - Dec. 2024

- Developed ramp-shaped RF-transparent antenna fairing to survive supersonic flight conditions.

Georgia Tech Motorsports (GTMS) - Aerodynamics Subsystem

Sept. 2022 - Sept. 2023

- Designed and manufactured MDF front wing molds and machined/sealed rear diffuser mold for layup and curing.

SKILLS

CAD/CAE: CATIA V5 | 3DX | Siemens NX | SolidWorks | Ansys | HyperWorks | Altium Designer

Programming: Python (Full Stack) | C/C++ | MATLAB | Java | JavaScript

Design/MFG: GD&T | Composite Design | Sheet Metal Design | Additive Manufacturing | PCB/Analog Circuit Design